

PROJECT REPORT

CUSTOMER CHURN & INTELLIGENCE DASHBOARD

Data Analytics Project

PostgreSQL · Python & Pandas · Power BI

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Table 1: Reading guide: how findings are labelled

Label	How it is used in this report
Data	Verified figures taken directly from the dataset and project outputs.
Observation	A pattern that is visible in the data (for example, the highest churn rate in a segment).
Interpretation	A cautious reading of what an observation may mean for the business. Not a causal claim.
Recommendation	A practical next step that follows from the findings.
Note	A caveat, definition or reading aid.

1 Executive Summary

Business problem. Customer churn removes recurring revenue from a subscription-based business. This project analyses 8,000 customer records to understand how churn is distributed across subscription types, locations, tenure, satisfaction, support behaviour and customer value, and to quantify the monthly revenue associated with customers who have churned.

Dataset and approach. The raw dataset contains 17 columns. It was inspected and validated with Python and Pandas (no missing values and no duplicate records were found), then extended with seven business-oriented engineered fields to give a 24-column analytical dataset. The data was stored in PostgreSQL for SQL business analysis, summarised into business KPIs, and presented in an interactive Power BI dashboard. The work is descriptive analytics; no predictive models were built.

8,000 Total Customers	2,505 Churned Customers	5,495 Active Customers	31.31% Churn Rate	264,401.10 Monthly Revenue at Risk
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Major findings.

- Overall churn is **31.31%**: 2,505 of 8,000 customers are classified as churned.
- Monthly spending associated with churned customers totals **264,401.10**. This is monthly revenue at risk, not lifetime revenue loss.
- Churn rates differ across segments, but the differences are relatively modest. Basic has the highest subscription churn (31.50%) and Florida the highest location churn (32.37%); the 25–36 month tenure band is highest by tenure (32.58%).
- Satisfaction alone does not clearly explain churn: the Excellent satisfaction category (32.72%) shows a higher observed churn rate than the Low category (29.88%).
- Basic accounts for 135,092.69 (51.09%) of monthly revenue at risk, which is in line with its 50.79% share of the customer base.

Dashboard outcome. The Power BI dashboard brings five KPI cards, five slicers and four charts into one executive-style view, allowing managers to explore churn and revenue at risk by segment. All findings in this report are observed patterns in the dataset; none of them establishes a causal relationship.

2 Introduction

2.1 What is customer churn?

Customer churn describes customers who stop using a company's product or service. In this dataset churn is recorded as a binary field: **Churn = 1** identifies a churned customer and **Churn = 0** identifies an active (non-churned) customer. The churn rate is the number of churned customers divided by the total number of customers.

2.2 Why churn analysis matters

In subscription-based businesses, revenue depends on customers continuing to pay over time. Each churned customer removes recurring revenue, and replacing that customer requires new acquisition. Measuring churn and understanding where it is concentrated gives management a factual basis for deciding where to look first, rather than treating all customers as one group.

2.3 Why customer intelligence is important

Customer intelligence turns raw customer records (demographics, subscription details, usage, support interactions and payments) into structured information that supports decisions. Business-oriented fields such as tenure bands, satisfaction levels and customer value make segments easier to compare and easier to explain to non-technical stakeholders.

2.4 How analytics supports retention

Descriptive analytics can show where churn is concentrated, how much revenue is associated with churned customers, and which segments deserve closer investigation. This project uses descriptive analytics only: it summarises observed patterns and does not build predictive models.

3

Business Problem

PROBLEM STATEMENT

The organisation needs to understand customer churn patterns across **subscription types, locations, tenure, satisfaction, support behaviour and financial value**, and to identify **where revenue is at risk**.

To address this, the project answers the following business questions:

- How many customers have churned, and what is the overall churn rate?
- Does churn differ across subscription types, locations, tenure bands and satisfaction levels?
- How do support activity, complaints and satisfaction scores compare between churned and active customers?
- How much monthly revenue is associated with churned customers, and in which segments is it concentrated?
- Which churned customers carry the highest customer value?
- How can these findings be delivered to decision-makers in an interactive form?

4 Project Objectives

Table 2: Project objectives and where they are addressed

#	Objective	Addressed in
1	Measure overall churn	Sections 9, 11
2	Understand churn across customer segments	Sections 9, 10, 12
3	Analyse customer behaviour	Sections 9, 10
4	Identify revenue associated with churned customers	Sections 11, 12
5	Analyse tenure and subscription patterns	Sections 9, 12
6	Analyse support and satisfaction behaviour	Sections 9, 12
7	Build business KPIs	Section 11
8	Create an interactive Power BI dashboard	Section 14
9	Provide actionable business recommendations	Section 13

5 Dataset Description

The source dataset (data/train.csv) contains **8,000 customer records** and **17 original columns**. After feature engineering, the analytical dataset (data/train_featured.csv) contains **24 columns**. Each row represents one customer, identified by a unique Customer_ID.

5.1 Churn definition

DEFINITION

Churn = 1: churned customer (2,505 customers). **Churn = 0:** active / non-churned customer (5,495 customers).

5.2 Engineered fields

Seven business-oriented fields were added during feature engineering: **Tenure_Category**, **Spending_Category**, **Usage_Intensity**, **Support_Intensity**, **Payment_Risk**, **Satisfaction_Level** and **Customer_Value**. They are described in Section 8.

5.3 Original variables

Table 3: Original dataset variables (17)

Variable	Type	Description
Customer_ID	Identifier	Unique customer identifier (8,000 distinct values)
Age	Integer	Customer age, range 18–69
Gender	Category	Female, Male
Location	Category	Illinois, New York, Texas, California, Florida
Subscription_Type	Category	Basic, Premium, Enterprise
Account_Age_Months	Integer	Length of the account in months, range 1–59
Monthly_Spending	Numeric	Monthly spend, range approximately 10.09–199.94
Total_Usage_Hours	Integer	Total usage hours, range 10–499
Support_Calls	Integer	Number of support calls, range 0–9
Late_Payments	Integer	Number of late payments, range 0–4
Streaming_Usage	Integer	Streaming usage measure, range 0–99
Discount_Used	Integer	Discount measure taking a broad range of values; it is not treated as a binary flag
Satisfaction_Score	Integer	Satisfaction score, range 1–10
Last_Interaction_Type	Category	Positive, Neutral, Negative
Complaint_Tickets	Integer	Number of complaint tickets, range 0–4
Promo_Opted_In	Binary	Promotional participation: 0 or 1
Churn	Binary (target)	1 = churned customer, 0 = active customer

6 Data Quality & Inspection

Data inspection was performed in Python with Pandas (`data_inspection.py`). The script reports shape, column names, sample records, data types, missing values, duplicates, unique-value counts, summary statistics, value ranges and categorical distributions.

6.1 Integrity checks

Table 4: Data integrity checks

Check	Result	Status
Row count	8,000	Pass
Column count (before feature engineering)	17	Pass
Missing values	0	Pass
Duplicate rows	0	Pass
Unique Customer_ID values	8,000 (one per row)	Pass
Active + churned = total	5,495 + 2,505 = 8,000	Pass

6.2 Range validation

Table 5: Verified value ranges of numeric variables

Variable	Minimum	Maximum
Age	18	69
Account_Age_Months	1	59
Monthly_Spending	≈ 10.09	≈ 199.94
Total_Usage_Hours	10	499
Support_Calls	0	9
Late_Payments	0	4
Streaming_Usage	0	99
Satisfaction_Score	1	10
Complaint_Tickets	0	4
Promo_Opted_In	0	1
Churn	0	1

6.3 Categorical validation

Table 6: Categorical value distributions

Variable	Category	Customers	Share
Gender	Female	4,035	50.44%
	Male	3,965	49.56%
Location	Illinois	1,624	20.30%
	New York	1,619	20.24%
	Texas	1,603	20.04%
	California	1,597	19.96%
	Florida	1,557	19.46%
Subscription Type	Basic	4,063	50.79%
	Premium	3,120	39.00%
	Enterprise	817	10.21%
Last Interaction Type	Neutral	2,700	33.75%
	Negative	2,650	33.12%
	Positive	2,650	33.12%

OBSERVATION

Gender, location and interaction type are close to evenly distributed. Subscription type is not: Basic represents roughly half of all customers, Premium 39.00% and Enterprise 10.21%. This matters when comparing revenue at risk across subscription types (Section 12).

7 Data Cleaning & Preparation

The preparation steps below were supported by the project scripts and results. The quality checks returned no missing values and no duplicate records, so no records were removed and no values needed to be imputed.

Table 7: Preparation steps performed

Step	Activity	Outcome
1. Load	Raw dataset read from data/train.csv with Pandas.	8,000 rows, 17 columns
2. Type validation	Data types of every column reviewed (df.dtypes).	Numeric, categorical and identifier fields confirmed
3. Missing-value check	Null counts calculated per column.	0 missing values
4. Duplicate check	Duplicate rows counted; Customer_ID uniqueness tested.	0 duplicates; 8,000 unique IDs
5. Range validation	Minimum and maximum reviewed for all numeric fields.	All values within plausible ranges (Section 6)
6. Categorical validation	Value counts reviewed for Gender, Location, Subscription_Type and Last_Interaction_Type.	Consistent category sets
7. Feature engineering	Seven business fields added and saved as data/train_featured.csv.	24 columns
8. Database preparation	A PostgreSQL table public.customer_churn was defined with one column per field, typed explicitly, with customer_id as primary key.	Dataset ready for SQL analysis

NOTE

Column names in the PostgreSQL table use lower-case snake_case (for example customer_id, monthly_spending), while the Pandas dataset uses capitalised names (for example Customer_ID, Monthly_Spending). This report uses the Pandas-style names when referring to fields.

8 Feature Engineering

Feature engineering added seven business-oriented fields to the original 17 columns. They translate raw numeric values into categories and measures that are easier to compare across segments and to present in a dashboard. The feature-engineering step is reported here by purpose and business meaning; the exact binning rules and formulas belong to the feature-engineering code and are not restated in this report.

Table 8: Engineered fields

Field	Data type (PostgreSQL)	Purpose	Business meaning and use
Tenure_Category	VARCHAR(20)	Groups account age into bands	Five 12-month bands (0–12, 13–24, 25–36, 37–48, 49–60 months). Used to compare churn across customer lifecycle stages.
Spending_Category	VARCHAR(20)	Groups monthly spending into categories	Allows churn to be compared across lower- and higher-spending customers.
Usage_Intensity	NUMERIC(10,2)	Summarises how heavily a customer uses the service	Compared between churned and active customers in the feature analysis.
Support_Intensity	INTEGER	Summarises support activity in a single measure	Used to analyse churn by support activity; a value of 8 or more flags “high support” customers in the SQL analysis.
Payment_Risk	VARCHAR(20)	Categorises payment behaviour	Allows churn to be compared across payment-risk groups.
Satisfaction_Level	VARCHAR(20)	Groups satisfaction score (1–10) into levels	Four levels: Low, Medium, Good, Excellent. Used in satisfaction analysis and as a dashboard slicer.
Customer_Value	NUMERIC(12,2)	Expresses the value of a customer in one number	Average of 3,165.31. A value of 5,000 or more defines “high-value” customers in the analysis, used to prioritise high-value churned customers.

NOTE

The thresholds quoted above ($\text{Support_Intensity} \geq 8$, $\text{Customer_Value} \geq 5,000$, $\text{Satisfaction_Score} \leq 3$) are the analytical cut-offs used in the SQL and Python analysis. They are analytical choices rather than properties of the source data.

9 Exploratory Data Analysis

Exploratory analysis was carried out with Pandas and Matplotlib (`feature_analysis.py`, `business_insights.py` and `visualization.py`). The sections below present the verified results. Differences are reported relative to the overall churn rate of 31.31%, in percentage points (pp).

NOTE

Several charts in this section use a vertical or horizontal axis that does not start at zero. This makes small differences look larger than they are. Please read the labelled values rather than bar lengths.

9.1 Churn distribution

Table 9: Class distribution of the target variable

Status	Customers	Share
Active (Churn = 0)	5,495	68.69%
Churned (Churn = 1)	2,505	31.31%
Total	8,000	100.00%

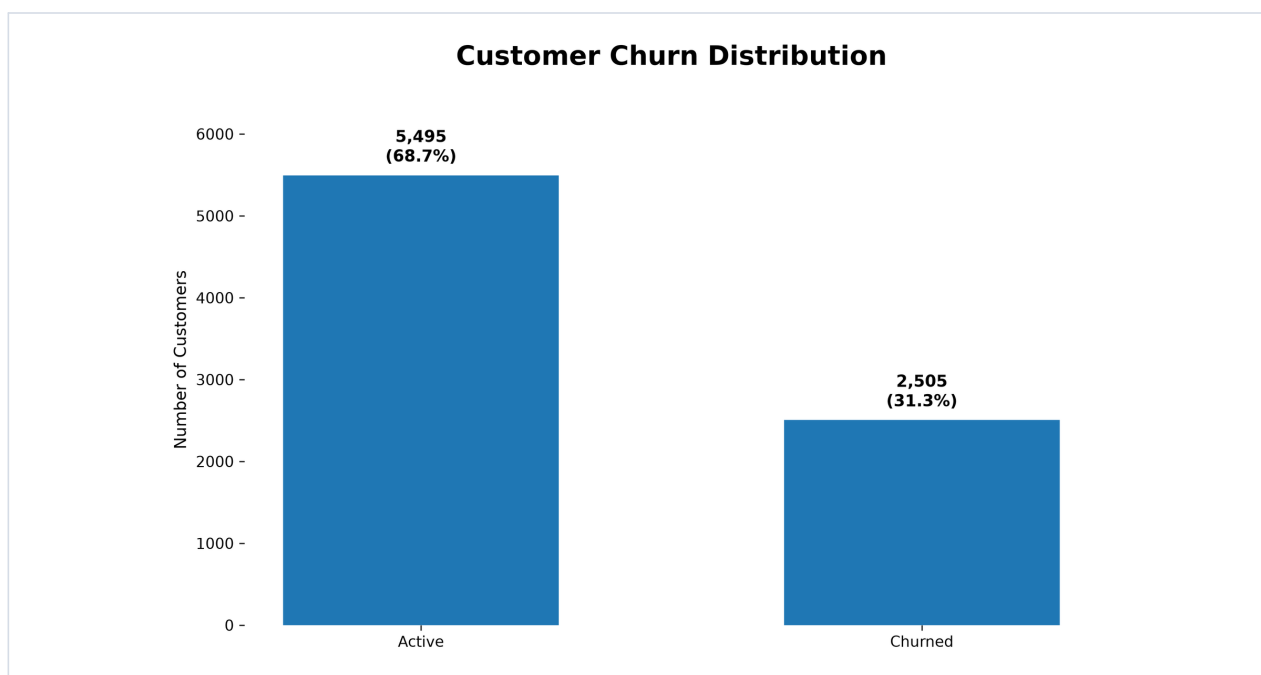


Figure 1: Customer churn distribution.

OBSERVATION

Almost one in three customers (31.31%) is classified as churned.

9.2 Subscription type

Table 10: Churn by subscription type

Subscription	Customers	Churned	Churn rate	Avg monthly spending
Basic	4,063	1,280	31.50%	105.49
Premium	3,120	970	31.09%	104.80
Enterprise	817	255	31.21%	101.42
All customers	8,000	2,505	31.31%	104.80

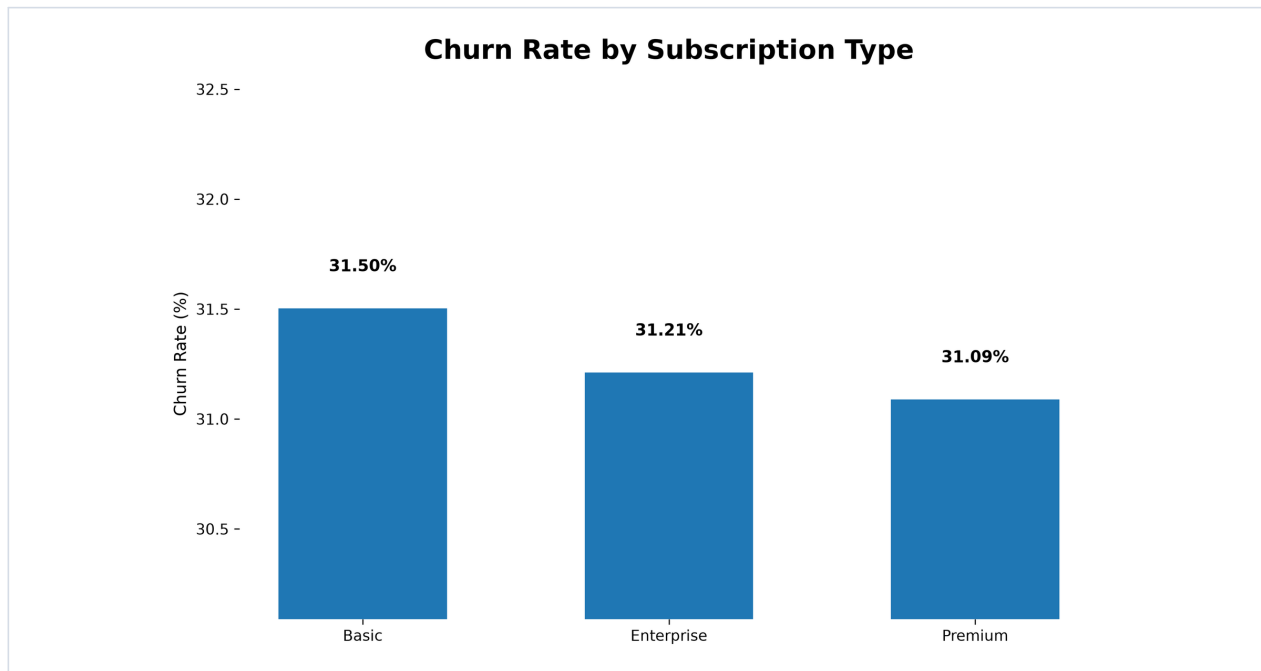


Figure 2: Churn rate by subscription type. *Y-axis does not start at zero.*

OBSERVATION

Basic has the highest observed churn rate (31.50%) and Premium the lowest (31.09%). The gap between the highest and lowest subscription type is only 0.41 pp.

9.3 Location

Table 11: Churn by location (ranked by churn rate)

Location	Customers	Churn rate	Difference vs overall (pp)
Florida	1,557	32.37%	+1.06
California	1,597	31.81%	+0.50
Texas	1,603	31.57%	+0.26
New York	1,619	31.44%	+0.13
Illinois	1,624	29.43%	-1.88

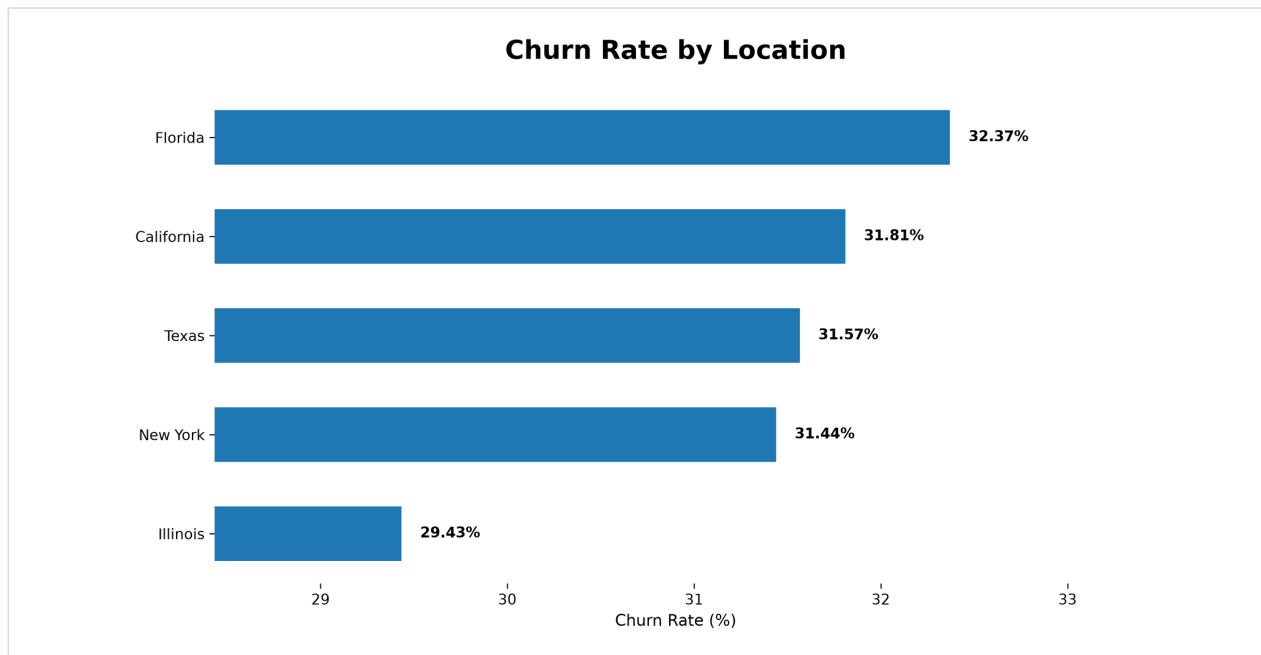


Figure 3: Churn rate by location. *X-axis does not start at zero.*

OBSERVATION

Florida has the highest observed churn rate (32.37%) and Illinois the lowest (29.43%), a spread of 2.94 pp. Illinois is the only location noticeably below the overall rate.

9.4 Tenure

Table 12: Churn by customer tenure

Tenure band	Churn rate	Difference vs overall (pp)
0–12 months	30.03%	–1.28
13–24 months	31.03%	–0.28
25–36 months	32.58%	+1.27
37–48 months	32.18%	+0.87
49–60 months	30.65%	–0.66

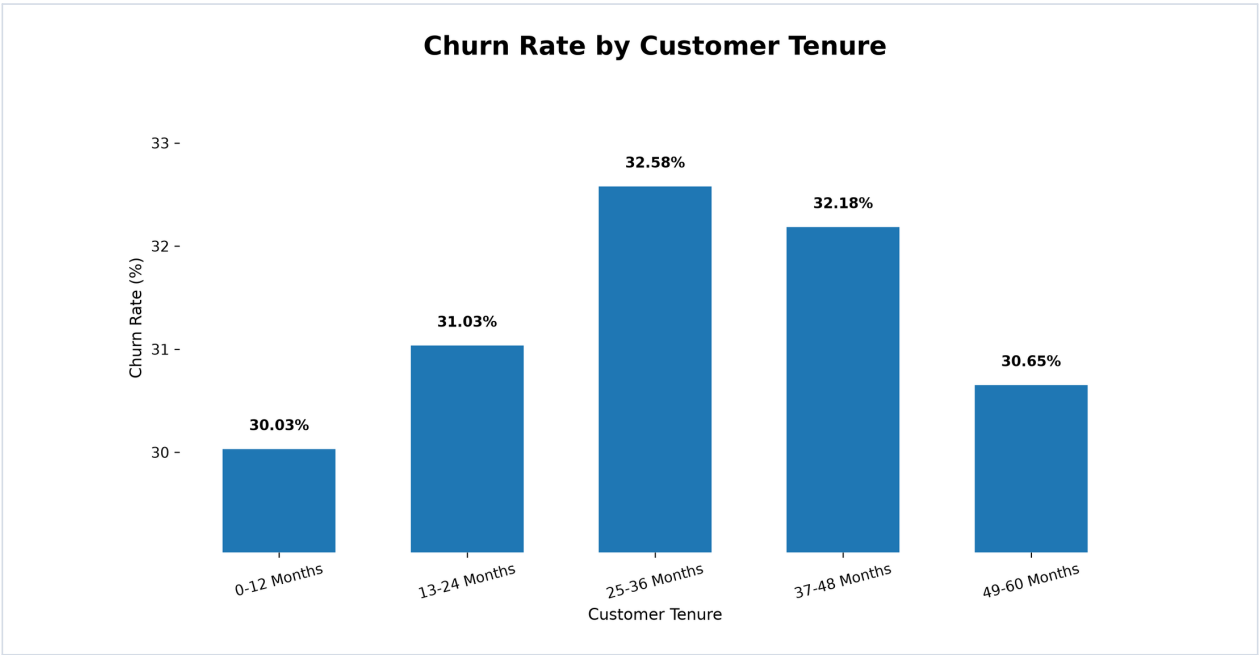


Figure 4: Churn rate by customer tenure. Y-axis does not start at zero.

OBSERVATION

Churn is highest in the 25–36 month band (32.58%) and the 37–48 month band (32.18%), and lowest in the 0–12 month band (30.03%). The pattern is not a steady rise or fall with tenure.

9.5 Satisfaction

Table 13: Churn by satisfaction level

Satisfaction level	Churn rate	Difference vs overall (pp)
Low	29.88%	−1.43
Medium	31.60%	+0.29
Good	31.71%	+0.40
Excellent	32.72%	+1.41

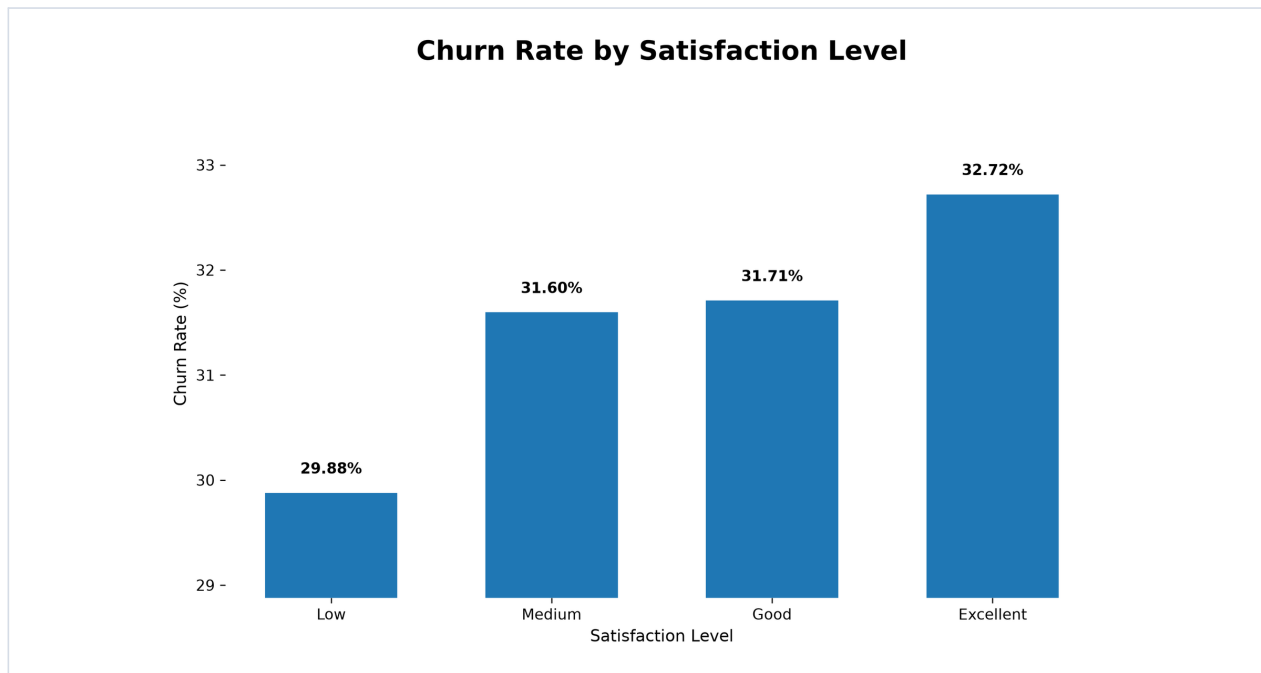


Figure 5: Churn rate by satisfaction level. *Y-axis does not start at zero.*

OBSERVATION

The Excellent satisfaction category (32.72%) has a higher observed churn rate than the Low category (29.88%).

INTERPRETATION

Satisfaction alone does not clearly explain churn in this dataset. The analysis does not establish a causal relationship between satisfaction and churn.

9.6 Support and complaints

Table 14: Support and satisfaction comparison by churn status

Measure (average)	Active customers	Churned customers	Difference (churned – active)
Support calls	4.48	4.40	–0.08
Complaint tickets	1.98	1.95	–0.03
Satisfaction score	5.42	5.55	+0.13

OBSERVATION

Average support calls, complaint tickets and satisfaction scores are very similar for churned and active customers. Churned customers have marginally fewer support calls and complaints and a marginally higher average satisfaction score.

INTERPRETATION

These variables are not shown to be causal drivers of churn, and the differences are small. Customer churn appears to be influenced by multiple factors rather than a single dominant variable.

9.7 Other dimensions examined

The SQL and Python analysis also examined churn by gender, last interaction type, spending category, payment risk, support intensity, complaint tickets, promotional participation and usage intensity. The verified result set supplied for this report includes the customer distributions of gender and interaction type (Section 6) but not the corresponding churn-rate breakdowns. Those breakdowns are therefore not tabulated here, and no pattern is claimed for them.

Table 15: Additional analyses carried out

Dimension	Analysis performed	Where
Gender	Churn rate by gender	SQL
Last interaction type	Churn rate by Positive / Neutral / Negative interaction	SQL
Spending category	Churn rate by Spending_Category	SQL, Python
Payment risk	Churn rate by Payment_Risk	SQL, Python
Support intensity and complaints	Churn rate by Support_Intensity and by Complaint_Tickets	SQL, Python
Promotional participation	Churn rate by Promo_Opted_In (0 / 1)	SQL
Usage intensity and customer value	Mean Usage_Intensity and Customer_Value for churned vs active customers	Python, SQL

10 SQL Analysis

SQL analysis was performed in PostgreSQL (`analysis.sql`) on the table `public.customer_churn`. The table holds all 24 fields of the featured dataset: categorical fields as `VARCHAR`, counts and scores as `INTEGER`, and monetary or intensity fields as `NUMERIC`. `customer_id` is the primary key.

Because Churn is stored as 0 or 1, the number of churned customers in any group is `SUM(churn)` and the churn rate is that sum divided by `COUNT(*)`.

10.1 Business questions answered with SQL

Table 16: Catalogue of SQL analyses

Theme	Business question	Techniques
Table checks	Does the table exist and how many rows does it hold? What do sample rows look like?	information_schema, COUNT, LIMIT
Overall KPIs	How many customers, churned customers, active customers; what is the churn rate, average spending and account age?	COUNT, SUM, AVG, ROUND
Segment churn	Churn by subscription, location, gender, last interaction type, tenure, spending category, payment risk and satisfaction level	GROUP BY, aggregates
Behaviour	Churn by support intensity, complaint tickets and promotional opt-in	GROUP BY, aggregates
Churned vs active	How do average spending, tenure, support calls, satisfaction and customer value differ by churn status?	GROUP BY churn, AVG
Revenue at risk	How much monthly spending belongs to churned customers, by subscription, location and tenure?	FILTER (WHERE), SUM
Risk concentration	Which subscription type generates the largest share of total revenue at risk?	CTE, window function
Priority customers	Which churned customers have the highest customer value, high support activity or low satisfaction?	WHERE, ORDER BY, LIMIT
Combined view	How does churn vary by location and satisfaction level together (groups of 50+ customers)?	GROUP BY, HAVING
Ranking	How do locations rank by churn rate?	CTE, RANK() OVER
Segmentation	Which risk segment does each customer belong to?	CASE

10.2 Representative SQL

The following excerpts are taken from the project's SQL file.

```
SELECT
    subscription_type,
    COUNT(*) AS total_customers,
    SUM(churn) AS churned_customers,
    ROUND(100.0 * SUM(churn) / COUNT(*), 2) AS churn_rate
FROM customer_churn
GROUP BY subscription_type
ORDER BY churn_rate DESC;
```

Listing 1: Churn rate by subscription type (aggregation and grouping)

```

WITH location_churn AS (
  SELECT
    location,
    COUNT(*) AS customers,
    SUM(churn) AS churned_customers,
    100.0 * SUM(churn) / COUNT(*) AS churn_rate
  FROM customer_churn
  GROUP BY location
)
SELECT
  location, customers, churned_customers,
  ROUND(churn_rate, 2) AS churn_rate,
  RANK() OVER (ORDER BY churn_rate DESC) AS churn_rank
FROM location_churn
ORDER BY churn_rank;

```

Listing 2: Ranking locations by churn rate (CTE and window function)

```

SELECT customer_id, subscription_type, monthly_spending, churn,
  CASE
    WHEN churn = 1 AND customer_value >= 5000 THEN 'High Value Churned'
    WHEN churn = 1 AND satisfaction_score <= 3 THEN 'Dissatisfied Churned'
    WHEN churn = 1 AND support_intensity >= 8 THEN 'High Support Churned'
    WHEN churn = 1 THEN 'Other Churned'
    ELSE 'Active'
  END AS customer_segment
FROM customer_churn;

```

Listing 3: Customer risk segmentation (CASE), abridged column list

10.3 SQL concepts used

Table 17: SQL concepts and their application

Concept	Use in this project
SELECT / FROM / ORDER BY / LIMIT	Table inspection, sorted result sets, top-20 high-value churned customers
WHERE	Filtering churned customers with high support activity or low satisfaction
GROUP BY with COUNT, SUM, AVG, ROUND	All segment-level churn tables
FILTER (WHERE churn = 1)	Revenue at risk: summing spending for churned customers only
HAVING	Location and satisfaction analysis restricted to groups of at least 50 customers
CASE	Customer risk segmentation
CTE (WITH)	Location ranking and revenue-at-risk share
Window functions: RANK() OVER, SUM() OVER ()	Ranking locations; percentage of total revenue at risk by subscription

10.4 Customer risk segmentation

The segmentation assigns each customer to one segment. Because the rules are evaluated in order, a churned customer who qualifies for more than one rule is placed in the first matching segment.

Table 18: Risk segmentation rules

Order	Segment	Rule
1	High Value Churned	Churn = 1 and Customer_Value $\geq$ 5,000
2	Dissatisfied Churned	Churn = 1 and Satisfaction_Score $\leq$ 3
3	High Support Churned	Churn = 1 and Support_Intensity $\geq$ 8
4	Other Churned	Any other churned customer
5	Active	Churn = 0

11 Business KPIs

The KPIs below were calculated with Pandas (`business_kpis.py`) and are the overall, unfiltered results for the full dataset of 8,000 customers.

8,000 Total Customers	2,505 Churned Customers	5,495 Active Customers	31.31% Churn Rate	264,401.10 Monthly Revenue at Risk
104.80 Avg Monthly Spending	3,165.31 Avg Customer Value	30.16 mo Avg Account Age	4.45 Avg Support Calls	5.46 Avg Satisfaction Score

Table 19: Overall business KPIs

KPI	Value	Definition
Total Customers	8,000	Unique Customer_ID values
Churned Customers	2,505	Customers with Churn = 1
Active Customers	5,495	Customers with Churn = 0
Churn Rate	31.31%	Churned Customers divided by Total Customers
Average Monthly Spending	104.80	Mean of Monthly_Spending
Average Customer Value	3,165.31	Mean of Customer_Value
Average Account Age	30.16 months	Mean of Account_Age_Months
Average Support Calls	4.45	Mean of Support_Calls
Average Satisfaction Score	5.46	Mean of Satisfaction_Score
Monthly Revenue at Risk	264,401.10	Sum of Monthly_Spending for customers with Churn = 1

DEFINITION

Monthly Revenue at Risk is the sum of monthly spending of customers who have churned. It is **not** lifetime revenue loss.

NOTE
Monetary values are reported exactly as they appear in the dataset; the currency unit is not specified in the source data.

12 Key Findings

The findings below are the strongest verified results of the analysis. Each is an **observation** about this dataset; none is a causal claim.

Table 20: Summary of key findings

#	Finding	Evidence
1	Overall churn is 31.31%.	2,505 of 8,000 customers
2	2,505 customers are classified as churned.	Churn = 1; 5,495 customers remain active
3	Monthly spending associated with churned customers totals 264,401.10.	Monthly revenue at risk
4	Basic has the highest observed subscription churn rate.	31.50%, versus 31.21% (Enterprise) and 31.09% (Premium)
5	Florida has the highest observed location churn rate.	32.37%; Illinois is lowest at 29.43%
6	The 25–36 month tenure group has the highest observed churn.	32.58%; the 37–48 month group follows at 32.18%
7	Satisfaction alone does not clearly explain churn.	Excellent (32.72%) is higher than Low (29.88%)
8	Revenue at risk is highest in Basic.	135,092.69, reflecting the size of the Basic customer base

12.1 Differences between segments are modest

Table 21: Spread of observed churn rates within each dimension (overall churn rate: 31.31%)

Dimension	Lowest	Highest	Spread (pp)
Subscription type	Premium 31.09%	Basic 31.50%	0.41
Location	Illinois 29.43%	Florida 32.37%	2.94
Tenure band	0–12 months 30.03%	25–36 months 32.58%	2.55
Satisfaction level	Low 29.88%	Excellent 32.72%	2.84

INTERPRETATION
Churn rates vary across customer segments, but the observed differences are relatively modest: no single dimension separates high-churn from low-churn customers by more than about three percentage points. This analysis did not include statistical significance testing, so small differences should not be over-interpreted.

12.2 Where monthly revenue at risk is concentrated

Table 22: Monthly revenue at risk by subscription type

Subscription	Share of customers	Share of churned customers	Monthly revenue at risk	Share of revenue at risk
Basic	50.79%	51.10%	135,092.69	51.09%
Premium	39.00%	38.72%	102,980.74	38.95%
Enterprise	10.21%	10.18%	26,327.67	9.96%
Total	100.00%	100.00%	264,401.10	100.00%

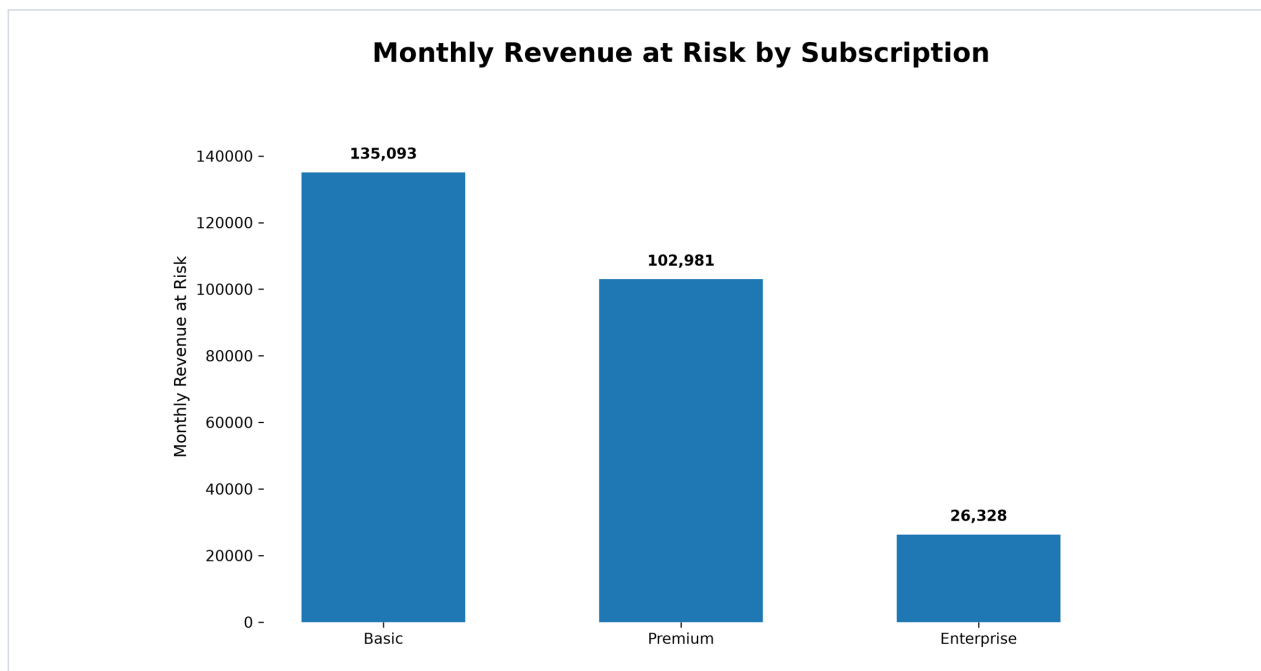


Figure 6: Monthly revenue at risk by subscription type.

OBSERVATION

Basic holds 51.09% of monthly revenue at risk, Premium 38.95% and Enterprise 9.96%. Each subscription type's share of revenue at risk is close to its share of the customer base (50.79%, 39.00%, 10.21%).

INTERPRETATION

Revenue risk is highest in the Basic segment mainly because Basic has the largest customer base and therefore the largest amount of churned-customer spending, not because its churn rate is markedly higher than that of the other subscription types.

13 Business Recommendations

The recommendations follow directly from the findings. Because observed differences are modest and descriptive, they are framed as areas to investigate and track rather than proven fixes.

Table 23: Recommendations and their basis

#	Recommendation	Basis in the findings
1	Prioritise retention analysis for Basic subscribers. Review where Basic churn and revenue at risk sit within the segment.	Highest subscription churn (31.50%) and largest revenue at risk (135,092.69; 51.09%)
2	Investigate customer-experience patterns in Florida, using Illinois as an internal comparison point.	Florida 32.37% versus Illinois 29.43%
3	Review retention opportunities in the 25–48 month tenure period.	25–36 months 32.58%; 37–48 months 32.18%
4	Use customer value to prioritise high-value churned or at-risk segments.	Customer_Value is available; SQL already flags high-value churned customers (value ≥ 5,000)
5	Combine multiple behavioural indicators (tenure, support activity, complaints, payment behaviour, usage, value) rather than relying on satisfaction alone.	Excellent satisfaction (32.72%) shows higher churn than Low (29.88%)
6	Track monthly revenue at risk as a recurring management KPI.	264,401.10 monthly spending associated with churned customers
7	Use the Power BI slicers for segment-level investigation, for example by combining location and subscription type.	Segment differences are modest, so combined views are needed to find concentrations
8	Validate before acting. Treat segment differences as hypotheses and test any retention action on a small scale before roll-out.	Observed differences are descriptive, not causal

RECOMMENDATION

Start with recommendations 1, 4 and 6. They rely on the clearest results in the data (where revenue at risk sits and how it can be tracked and prioritised), whereas the location and tenure differences are small and are best treated as areas to investigate.

14 Power BI Dashboard

The final deliverable is an interactive Power BI dashboard titled **CUSTOMER CHURN & INTELLIGENCE DASHBOARD**. It lets managers explore churn and revenue at risk without writing queries.

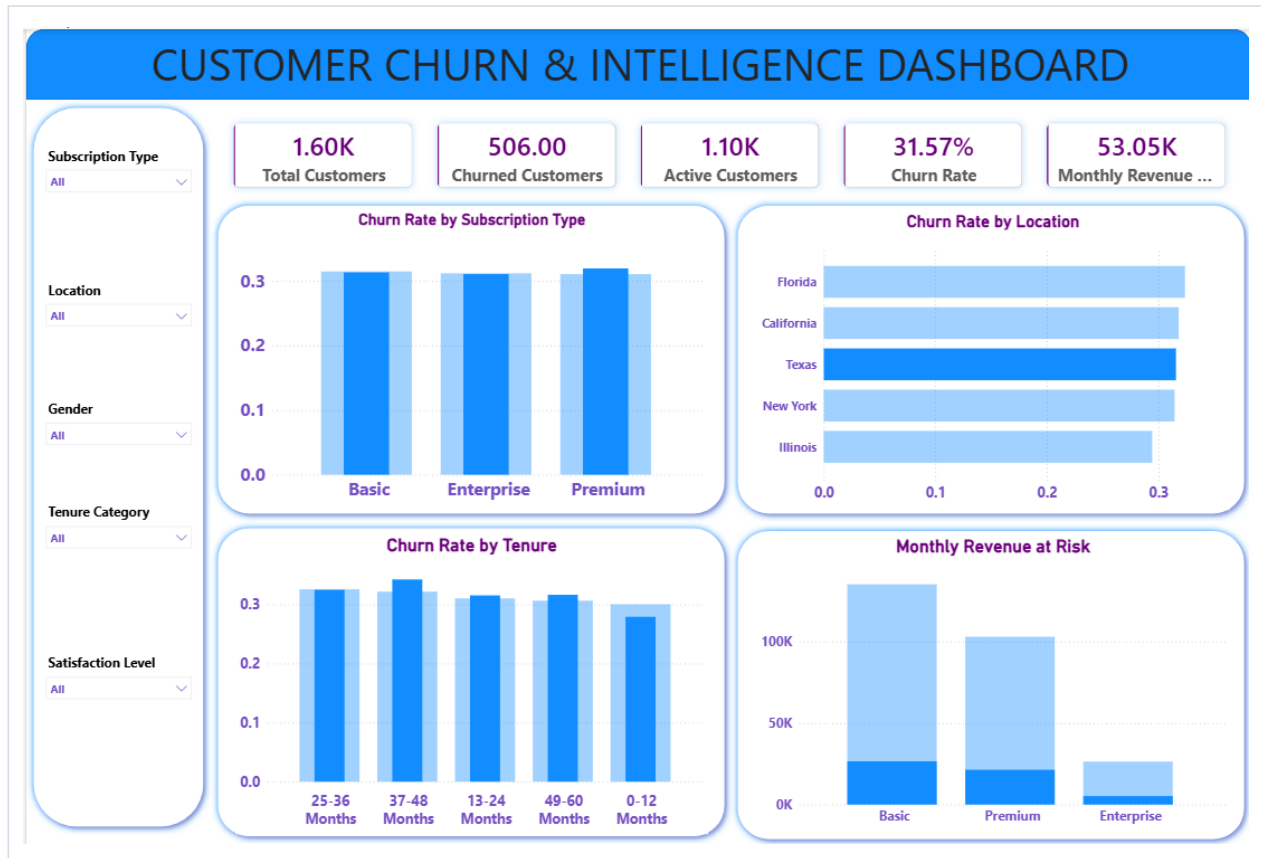


Figure 7: Customer Churn & Intelligence Dashboard (Power BI).

NOTE

The screenshot shows the dashboard in a filtered state: all five slicers read "All", but Texas is highlighted in the location chart. The KPI cards therefore show Texas figures (1.60K customers, 506 churned, 1.10K active, 31.57% churn, 53.05K monthly revenue at risk), consistent with Texas's 1,603 customers and 31.57% churn rate. The overall project KPIs are those in Section 11.

14.1 Layout

The dashboard uses an executive-style layout with a blue primary theme, rounded visual containers and a clear visual hierarchy: title banner at the top, filter panel on the left, KPI cards in a row beneath the title, and four charts arranged in a two-by-two grid.

Table 24: Dashboard components

Component	Content
KPI cards (5)	Total Customers, Churned Customers, Active Customers, Churn Rate, Monthly Revenue at Risk
Slicers (5)	Subscription Type, Location, Gender, Tenure Category, Satisfaction Level (left-side filter panel)
Chart 1	Churn Rate by Subscription Type (Basic, Enterprise, Premium)
Chart 2	Churn Rate by Location (Florida, California, Texas, New York, Illinois)
Chart 3	Churn Rate by Tenure (five 12-month tenure bands)
Chart 4	Monthly Revenue at Risk by Subscription (Basic, Premium, Enterprise)

14.2 Interactivity and business usability

- **Slicers** filter every KPI card and chart at once, so any segment combination (for example Basic customers in Florida) can be examined.
- **Cross-highlighting:** selecting a bar in one chart highlights the matching portion in the other charts, shown in the dashboard as a darker blue portion within each lighter total bar.
- **At-a-glance monitoring:** the KPI row gives the headline position of the current selection.
- **Prioritisation:** the revenue-at-risk chart shows where churned-customer spending is concentrated.
- **Consistent definitions:** the KPIs match those calculated in SQL and Python, with revenue at risk defined as monthly spending of churned customers.

15 Technology Stack

Table 25: Technologies used

Technology	Role in this project
PostgreSQL	Database and SQL analysis (table public.customer_churn)
Python	Data inspection, cleaning checks, feature engineering and exploratory analysis
Pandas	Data manipulation, aggregation and KPI calculation
Matplotlib	Exploratory charts (churn distribution, subscription, tenure, satisfaction, location, revenue at risk)
Power BI	Interactive business dashboard
DAX	KPI measures in the dashboard

15.1 Project files

Table 26: Project files and their purpose

File	Purpose
<code>data_inspection.py</code>	Shape, columns, types, missing values, duplicates, unique values, ranges, categories
<code>feature_analysis.py</code>	Churn by engineered fields; customer value and usage intensity by churn status
<code>business_kpis.py</code>	Overall business KPIs
<code>business_insights.py</code>	Subscription, location, tenure and satisfaction analysis; revenue at risk; high-value churned customers
<code>visualization.py</code>	Matplotlib charts exported as PNG files
<code>analysis.sql</code>	PostgreSQL table definition and SQL business analysis

16 Project Workflow

The project follows a linear workflow from raw data to business recommendations.



Figure 8: End-to-end project workflow.

17 Limitations

- **The analysis is descriptive.** It summarises observed patterns and does not model or predict churn.
- **The dataset does not establish causality.** A difference in churn rate between segments does not show that the segment attribute caused the churn.
- **Observed differences are modest.** Many segment differences are only one to three percentage points, and no statistical significance testing was performed.
- **Monthly revenue at risk is not lifetime revenue loss.** It is the monthly spending of churned customers only.
- **The dataset is a specific analytical sample.** Findings should not automatically be generalised to all customers or businesses.
- **No time dimension is available.** The dataset is a single snapshot, so churn trends over time and the timing of churn cannot be analysed.
- **Analytical categories are choices.** Bands and thresholds such as tenure bands, satisfaction levels and the high-value cut-off affect how segments look.
- **Chart axes.** Several exploratory charts use truncated axes, which visually amplify small differences; the labelled values should be used for interpretation.

18 Conclusion

This project demonstrates an end-to-end Data Analytics workflow, from raw customer data to SQL analysis, Python exploratory analysis, KPI development and interactive Power BI reporting. An 8,000-record dataset was validated (no missing values, no duplicates, 8,000 unique customers), enriched with seven business-oriented fields, stored in PostgreSQL and analysed with a structured set of business queries.

The analysis shows an overall churn rate of 31.31% (2,505 customers) and 264,401.10 in monthly spending associated with churned customers. Churn rates differ across subscription types, locations, tenure bands and satisfaction levels, but the differences are relatively modest, and satisfaction alone does not clearly explain churn. The largest concentration of monthly revenue at risk is in the Basic subscription segment, in line with its share of the customer base.

The Power BI dashboard packages these results into five KPI cards, five slicers and four charts so that decision-makers can explore segments themselves. The findings are descriptive; they identify where to look and what to track, and they should be validated before any retention action is taken.

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Future Scope

Table 27: Possible analytics extensions

Extension	Value
Customer-level retention tracking	Follow individual customers over time instead of a single snapshot
Cohort analysis	Compare groups of customers by start period to see how retention develops
Time-series churn trends	Monitor churn month by month, if historical data becomes available
Customer lifetime value segmentation	Extend Customer_Value into richer value-based segments
Automated dashboard refresh	Keep KPIs and revenue at risk current without manual updates
More granular behavioural analysis	Examine combinations of tenure, usage, support and payment behaviour
CRM and support data integration	Add interaction history and reasons for contact to the customer view

NOTE

Predictive modelling would be a possible longer-term extension beyond the scope of this descriptive analytics project. It is not part of the work reported here.